

NORTHVALE THERAPEUTICS | MEDICAL AFFAIRS

Obesity: Disease Burden, Pathophysiology and Management

An evidence overview for internal scientific exchange

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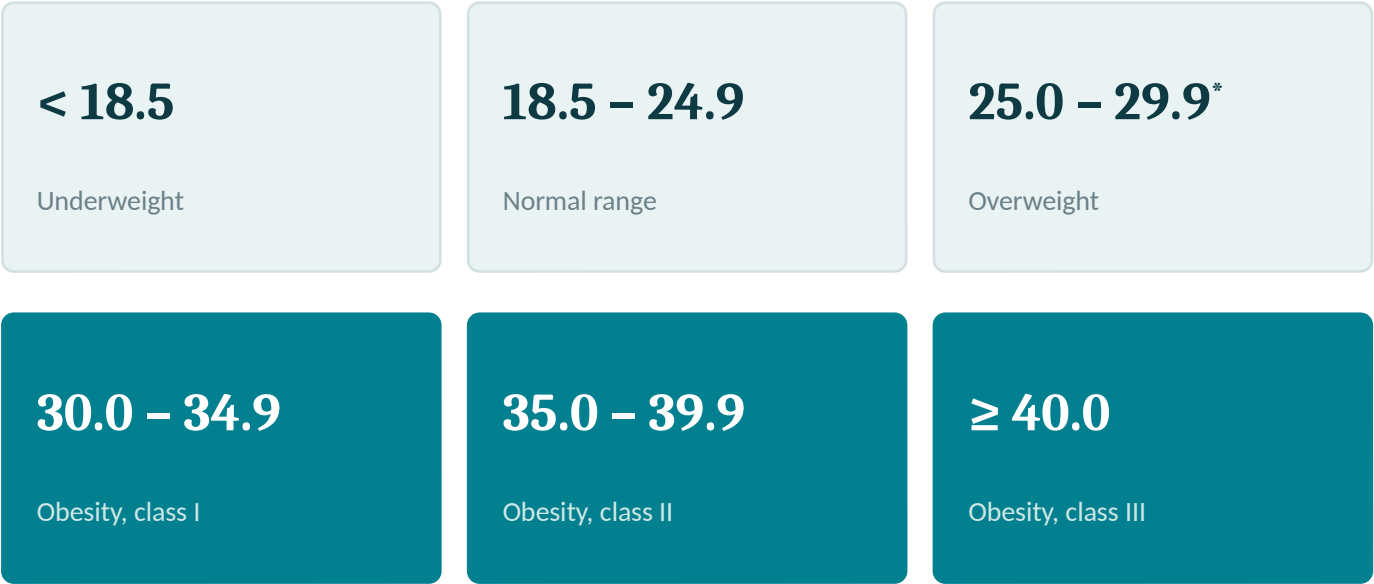
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Defining and classifying obesity

Obesity is characterised by excess adipose accumulation sufficient to impair health¹⁻³. Body mass index remains the primary population screening metric, calculated as weight in kilograms divided by height in metres squared.

BMI does not distinguish adipose tissue from lean mass, and performs inconsistently across ancestry groups and at the extremes of stature^{2,4}. Waist circumference and waist-to-height ratio are recommended as adjunct measures⁵.



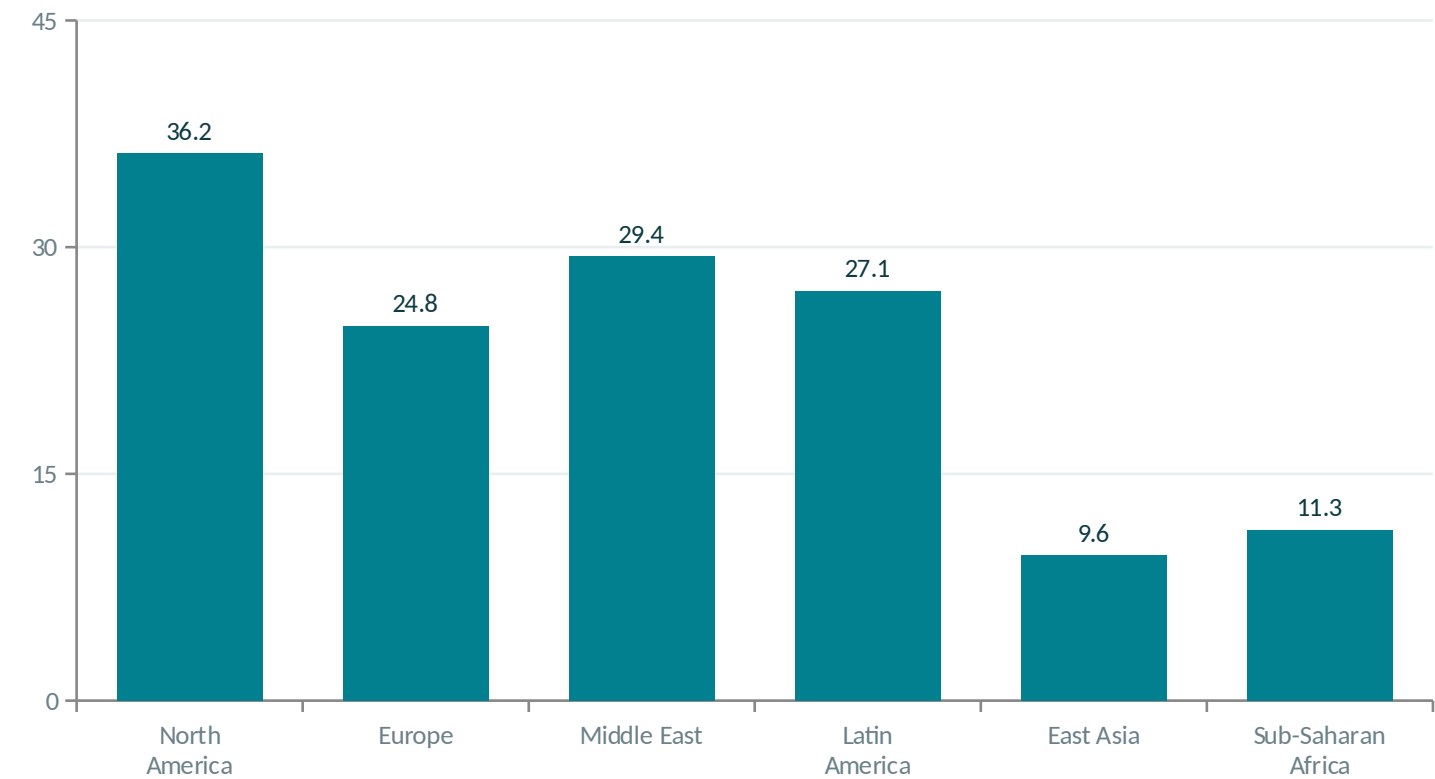
BMI, kg/m²

* The overweight band is not ancestry-adjusted; lower cut-offs apply in some populations⁵.

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Prevalence is rising across every region



Key observations

- Age-standardised prevalence has approximately doubled in adults since 1990 across most reporting regions⁶⁻⁸.
- Modelling projects 1.9 billion adults above the class I threshold by 2035 under current trajectories⁹.
- Urban incidence exceeds rural incidence in high-income settings, with the gradient reversing in several low-income settings^{10, 11}.
- Regional figures conceal substantial within-country variation by income decile^{6,9-11}.

References

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Energy homeostasis and adipose expansion

01

Energy surplus

Sustained positive energy balance drives triglyceride storage in subcutaneous depots¹².

02

Hypothalamic signalling

Leptin and insulin afferent signalling attenuates, raising the defended body-weight set point^{12,13}.

03

Adipocyte hypertrophy

Storage capacity is exceeded; lipid deposits ectopically in liver and skeletal muscle¹⁴.

04

Metabolic dysregulation

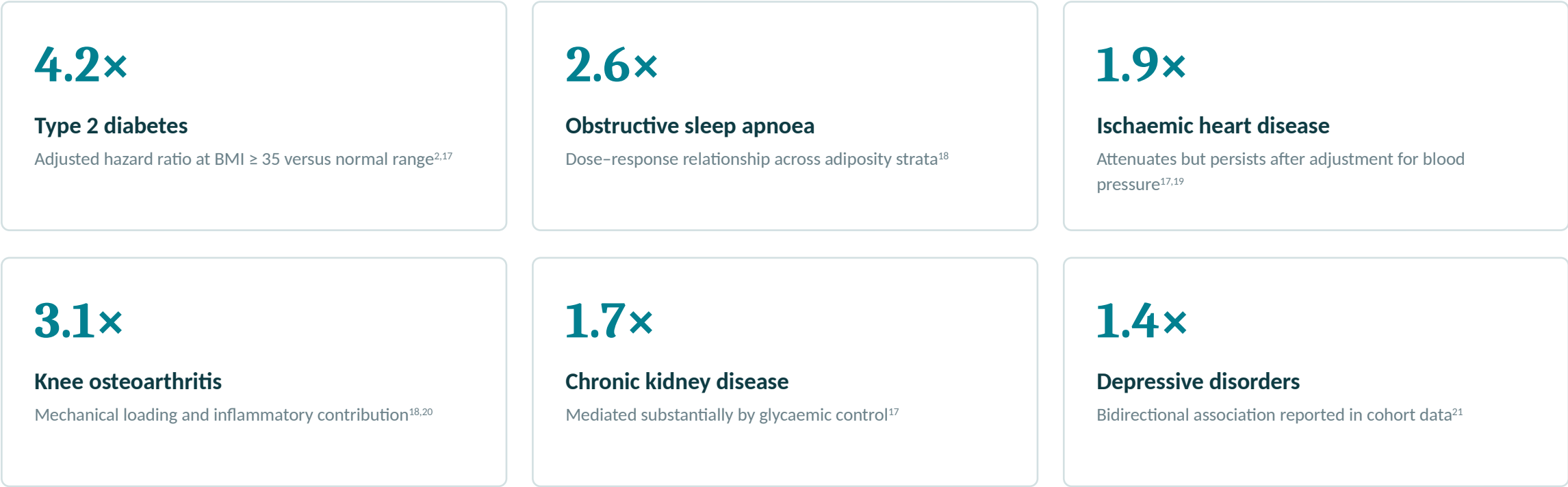
Local inflammation and insulin resistance establish a self-reinforcing cycle¹⁴⁻¹⁶.

The defended set point explains why weight regain follows energy restriction in the majority of participants^{12,13}.

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Associated conditions and relative risk



Relative risks are illustrative and drawn from pooled cohort estimates; confidence intervals are omitted for clarity.

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Direct and indirect cost burden

\$173bn

Annual direct medical cost

Attributable spend across inpatient, outpatient and pharmacy settings²²

2.4%

Share of national health expenditure

Median across twelve high-income reporting systems²²

\$89bn

Indirect productivity loss

Absenteeism, presenteeism and premature workforce exit²³

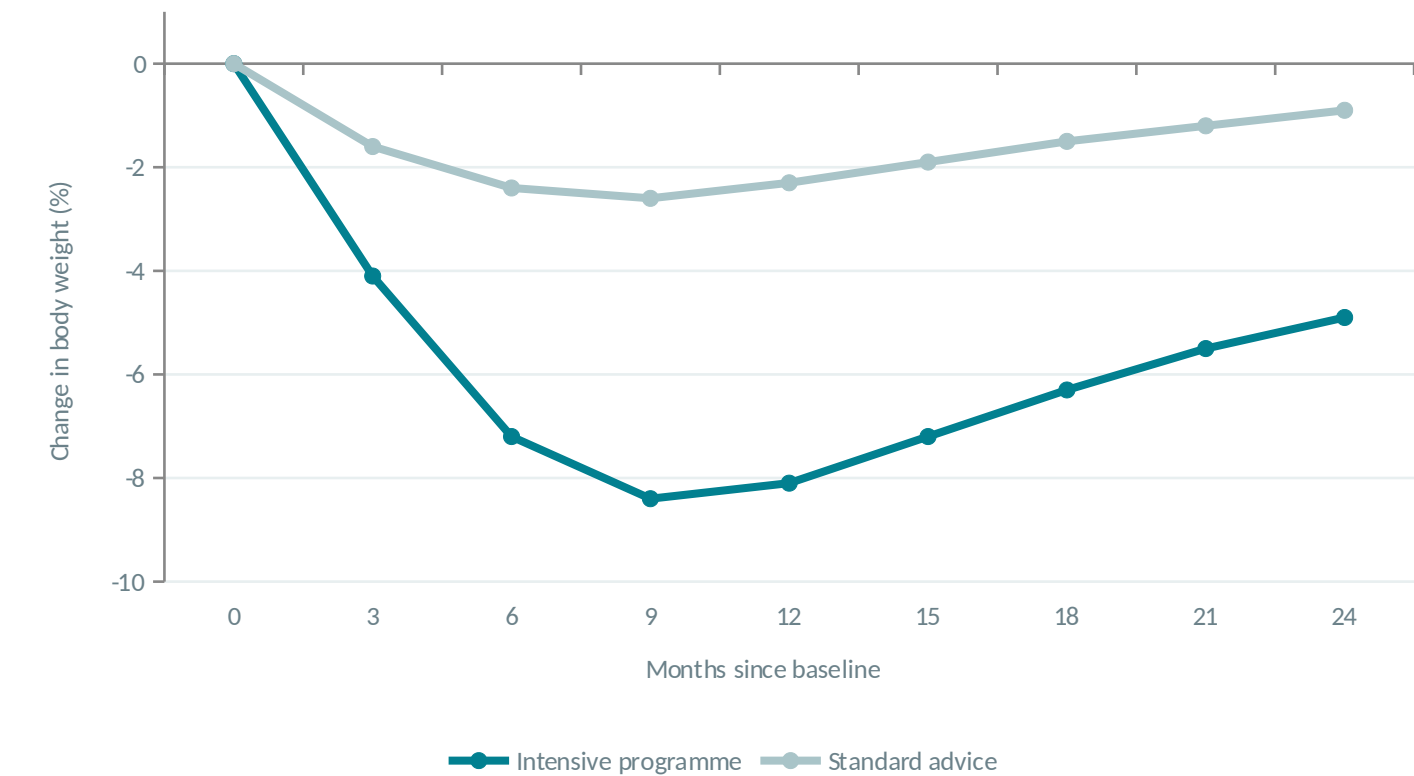
Cost attribution methodology varies materially between published estimates. Prevalence-based approaches and incidence-based approaches produce figures that differ by a factor of two or more for the same population, and cross-study comparison is therefore inadvisable without harmonisation^{22,23}.

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Structured lifestyle intervention at 24 months



Interpretation

- Nadir weight loss occurs at month nine in the intensive arm, with partial regain thereafter²⁴.
- Approximately 60% of maximal loss is retained at 24 months among programme completers^{24,25}.
- Attrition reaches 34% by month 18 and is the principal limitation of the evidence base²⁵.
- Glycaemic benefit persists beyond the weight trajectory in participants with prediabetes^{17,24}.

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Mechanistic classes in weight management

Incretin pathway

- Mean reduction 12–18% at 68 weeks²⁶
- Gastrointestinal events most common²⁷
- Discontinuation 7–14% in trial settings^{26,27}

Central appetite modulation

- Mean reduction 5–9% at 52 weeks²⁶
- Neuropsychiatric monitoring required²⁷
- Contraindicated in uncontrolled hypertension²⁶

Absorption inhibition

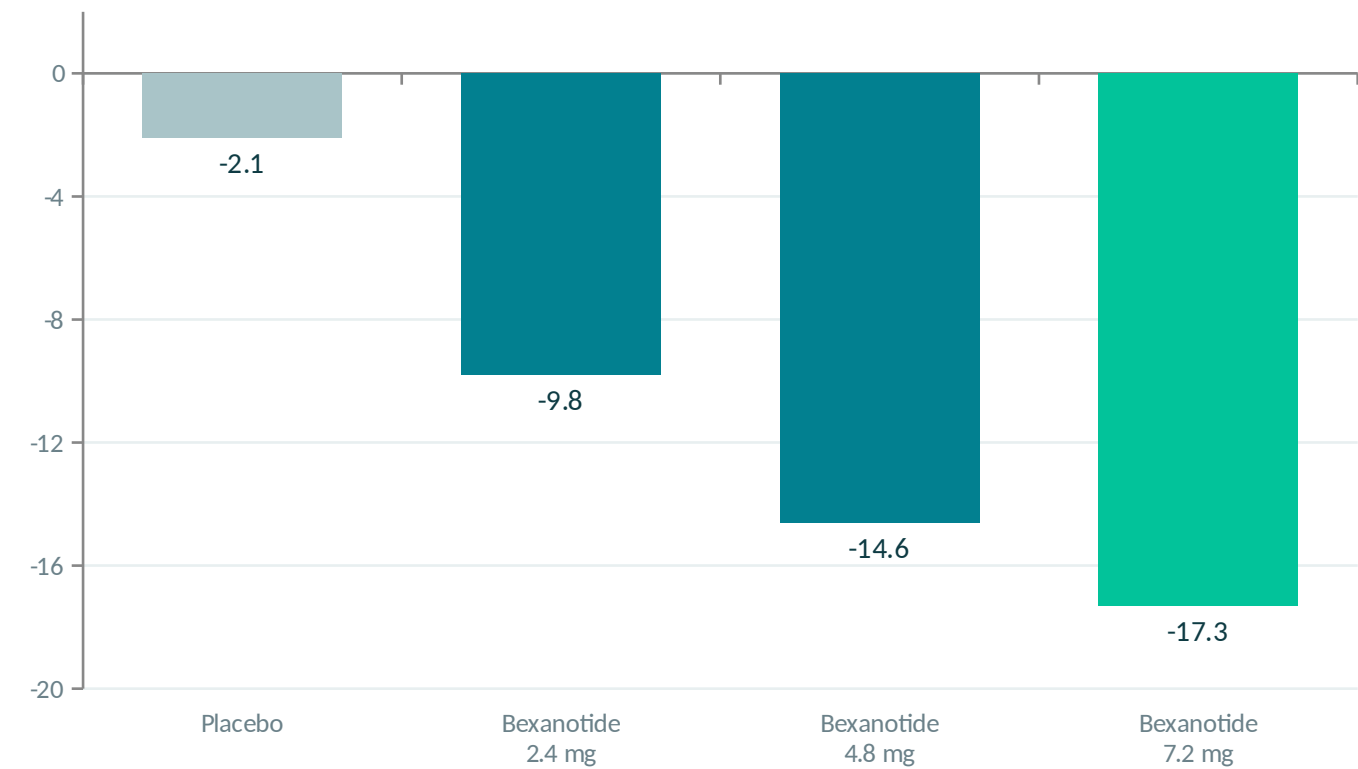
- Mean reduction 3–5% at 52 weeks²⁶
- Steatorrhoea limits adherence²⁸
- Fat-soluble vitamin supplementation advised^{26,28}

Adherence at 12 months remains below 45% across all three classes in real-world cohorts, and is the dominant determinant of realised benefit²⁶⁻²⁸.

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ZENITH-3: primary endpoint at week 68



Trial design

- Randomised, double-blind, placebo-controlled, 68 weeks; n = 2,140 across 94 sites²⁹.
- Adults with BMI ≥ 30 , or ≥ 27 with at least one weight-related comorbidity²⁹.
- All active arms met the primary endpoint versus placebo ($p < 0.001$)³⁰.
- Dose-response was preserved across prespecified subgroups^{29,30}.

Bexanotide is a fictional investigational compound. These data are invented for testing purposes.

References

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Guidance and practice implications

1

Stage, do not just measure

Classification should combine anthropometry with functional and metabolic staging rather than BMI alone³¹.

2

Escalate on response

Treatment intensity should follow documented response at defined review points rather than baseline BMI^{31,32}.

3

Plan for chronicity

Weight regain after intervention withdrawal is the expected course; follow-up should extend beyond 24 months³³.

4

Document the pathway

Local care pathways should record the rationale for each escalation step to support later audit^{26,31-34}.

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